

TEST 3: (n^{th} Root Test)

Given an infinite series $\sum_{n=1}^{\infty} U_n$, where U_n - Positive terms.

n^{th} Root test (states)

* If $\lim_{n \rightarrow \infty} (U_n)^{\frac{1}{n}} = p < 1$, then $\sum_{n=1}^{\infty} U_n$ is Convergent

* If $\lim_{n \rightarrow \infty} (U_n)^{\frac{1}{n}} = p > 1$, then $\sum_{n=1}^{\infty} U_n$ is divergent.

* If $\lim_{n \rightarrow \infty} (U_n)^{\frac{1}{n}} = p = 1$, then root test is inconclusive.

Example 1

Use root test to determine whether the following series are Convergent or divergent.

(a) $\sum_{n=1}^{\infty} \frac{n^2}{2^n}$

(b) $\sum_{n=1}^{\infty} \left(1 - \frac{1}{3n}\right)^{n^2}$

(c) $\sum_{n=1}^{\infty} \frac{2^n}{n^3}$

(d) $\sum_{n=1}^{\infty} \left(\frac{1}{1+n}\right)^n$

Solution

(a) $\lim_{n \rightarrow \infty} \left(\frac{n^2}{2^n}\right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{(\sqrt[n]{n})^2}{2} = \frac{1}{2} < 1 \quad \therefore \sum_{n=1}^{\infty} \frac{n^2}{2^n}$ is Convergent.

(b) $\lim_{n \rightarrow \infty} \left[\left(1 - \frac{1}{3n}\right)^{n^2}\right]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(1 - \frac{1}{3n}\right)^n = e^{-\frac{1}{3}} = \frac{1}{\sqrt[3]{e}} < 1 \quad \therefore$ Convergent

(c) $\lim_{n \rightarrow \infty} \left(\frac{2^n}{n^3}\right)^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \frac{2}{(\sqrt[n]{n})^3} = 2 > 1 \quad \therefore$ divergent

(d) $\lim_{n \rightarrow \infty} \left[\left(\frac{1}{1+n}\right)^n\right]^{\frac{1}{n}} = \lim_{n \rightarrow \infty} \left(\frac{1}{n+1}\right) = 0 < 1 \quad \therefore$ Convergent

Test 4: Integral Test

Given $\sum_{n=1}^{\infty} u_n = \sum_{n=1}^{\infty} f(n) \leftarrow \text{positive terms and decreasing function}$

* If $\int_1^{\infty} f(x) dx$ Converges then $\sum_{n=1}^{\infty} u_n$ Converges

* If $\int_1^{\infty} f(x) dx$ diverges then $\sum_{n=1}^{\infty} u_n$ also diverges.

THE P-Series

$\sum_{n=1}^{\infty} \frac{1}{n^p}$ is called p-series, using integral test we have

$$\sum_{n=1}^{\infty} \frac{1}{n^p} = \begin{cases} \text{Converges if } p > 1 \\ \text{diverges if } p \leq 1 \end{cases}$$

Example 2

Use integral test to determine convergence/divergence of

(a) $\sum_{n=1}^{\infty} \frac{1}{n^2+1}$

(b) $\sum_{n=1}^{\infty} \frac{1}{n^2}$

(c) $\sum_{n=1}^{\infty} \frac{n}{n^2+4}$

Solution

(a) $\int_1^{\infty} \frac{1}{x^2+1} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{1}{x^2+1} dx = \lim_{b \rightarrow \infty} [\tan^{-1} x]_1^b = \lim_{b \rightarrow \infty} [\tan^{-1} b - \frac{\pi}{4}] = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$
 $\therefore \sum_{n=1}^{\infty} \frac{1}{n^2+1}$ is Convergent.

(b) $\sum_{n=1}^{\infty} \frac{1}{n^2}$, p-series with $p=2 > 1 \therefore$ Convergent

(c) $\int_1^{\infty} \frac{x}{x^2+4} dx = \lim_{b \rightarrow \infty} \int_1^b \frac{x}{x^2+4} dx = \lim_{b \rightarrow \infty} \left[\frac{1}{2} \ln(x^2+4) \right]_1^b = \lim_{b \rightarrow \infty} \frac{1}{2} [\ln(b^2+4) - \ln 5] = \infty$
diverges

$\therefore \sum_{n=1}^{\infty} \frac{n}{n^2+4}$ is divergent.